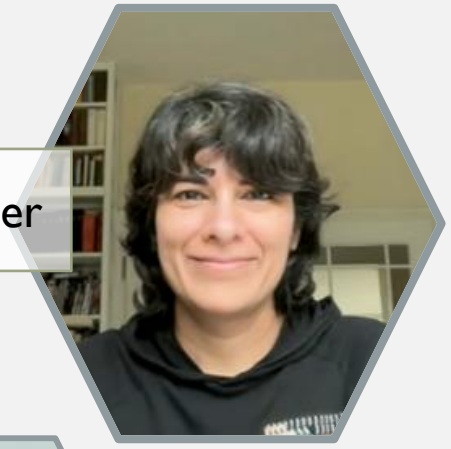
DECARBONISATION DYNAMICS IN THE EUROPEAN STEEL INDUSTRY

INTEGRATING
📄 CORPORATE CLIMATE NARRATIVES,
📈 EMISSION TRAJECTORIES, AND
⚡ EXTERNAL SHOCKS

15 DECEMBER 2025

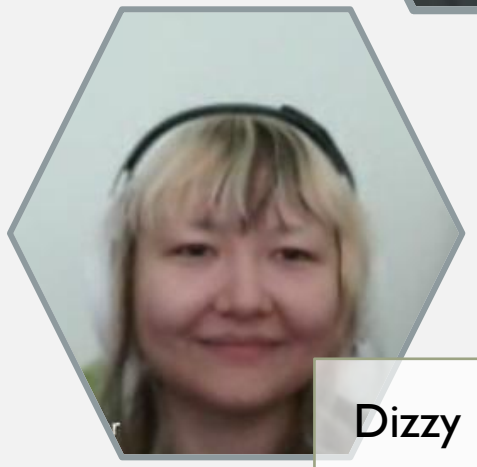
OUR TEAM 🔗 🌱



Maryam J Rutner

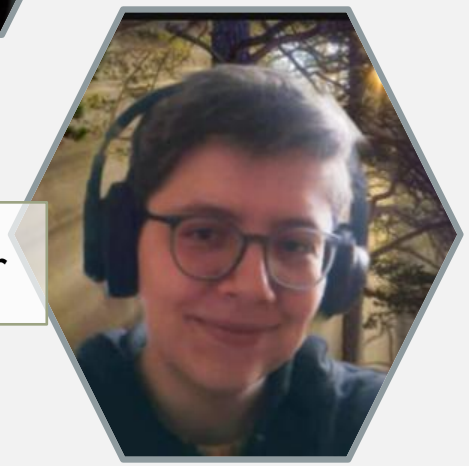
Apostolos Koumparos

SuSteelAible



Dizzy Englmaier

Karl Rackwitz



Irene Polgar

THE STEEL SECTOR…

7-9% of global CO_2 emissions

one of the most **carbon-intensive** industrial sectors

regulatory and investor **pressure** (Green Deal)

urgency to decarbonize

TWO-FOLD APPROACH

„ACTION“

„TALK“

CONCEPT

External drivers

Internal barriers & motivators

ANALYSIS

Trend analysis

OLS/ Panel regression

NLP

ClimateBERT, RAG, OPUS-MT

OUTPUT

TRANSITION TRACKER

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DATA LANDSCAPE

- 15 steel producing companies in the EU
- 200+ reports for years 2013-2024

company	year	CO_2 e / t steel
Outokumpu	2024	0.48
Celsa	2024	0.24
SSAB	2024	1.46
ArcelorMittal	2024	1.87
…		

TARGET

- CO_2 e /production
- Year
- Company

+

year	carbon price	electricity price	EU parl change
2024	65	0.1913	1
2023	84	0.2077	
2022	80.32	0.1972	
2021	52.[illegible]		
…	…		

EXTERNAL

- Carbon price
- Technology
- Policy changes
- …

+

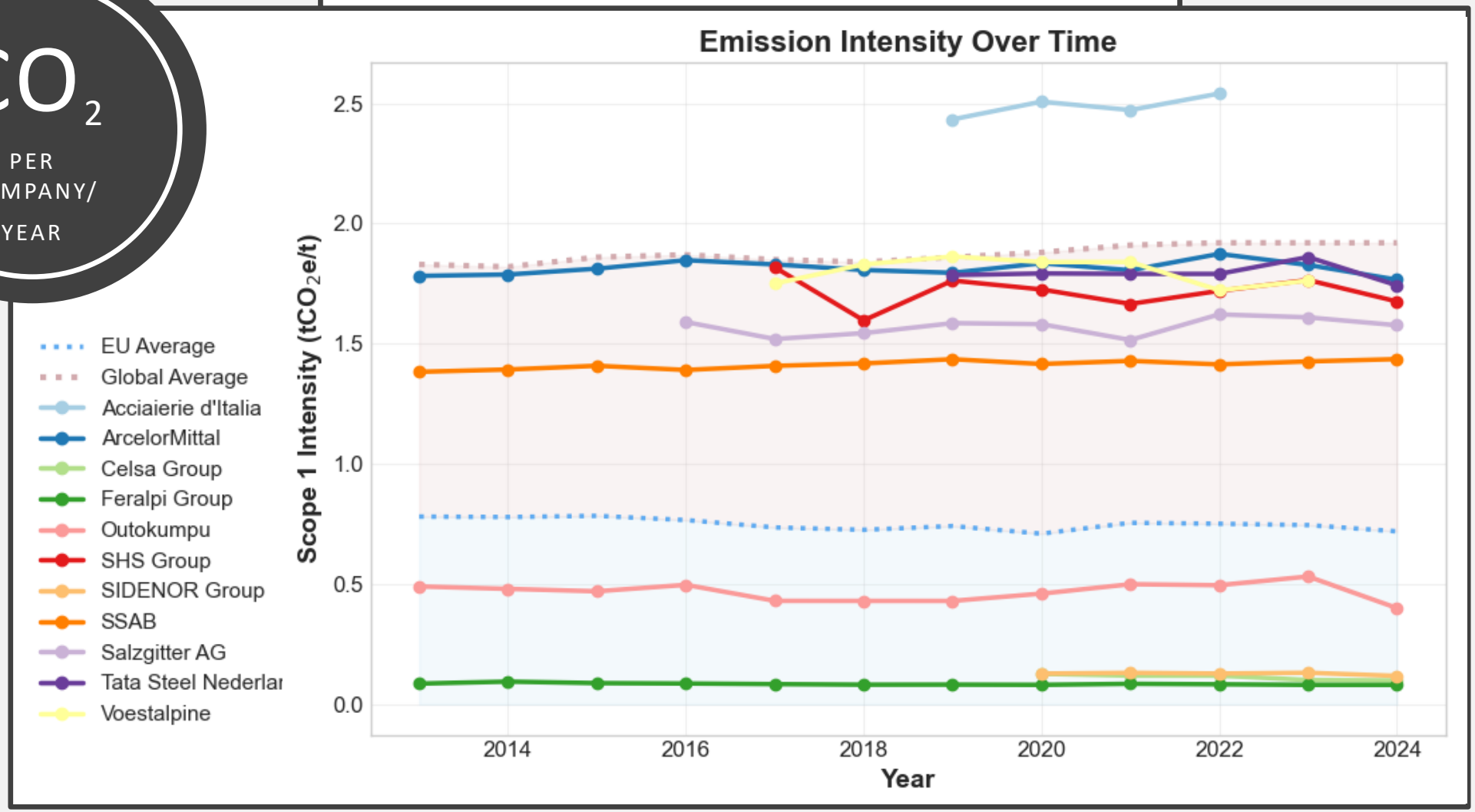
INTERNAL

- Sustainability reports
- Annual reports
- Varying availability

ACTION – OR NO ACTION?

CO_2

PER COMPANY/ YEAR



- No significant changes in trend detectable

WHAT INFLUENCES EMISSION INTENSITY?

Policies

Regulatory mandates and climate commitments

Paris Agreement
Green Deal
CBAM
IED

Pricing

Fuel & Materials

Carbon
Electricity
Gas
Coal
Iron Ore

ETS

EU Emissions Trading System
Cap & Trade

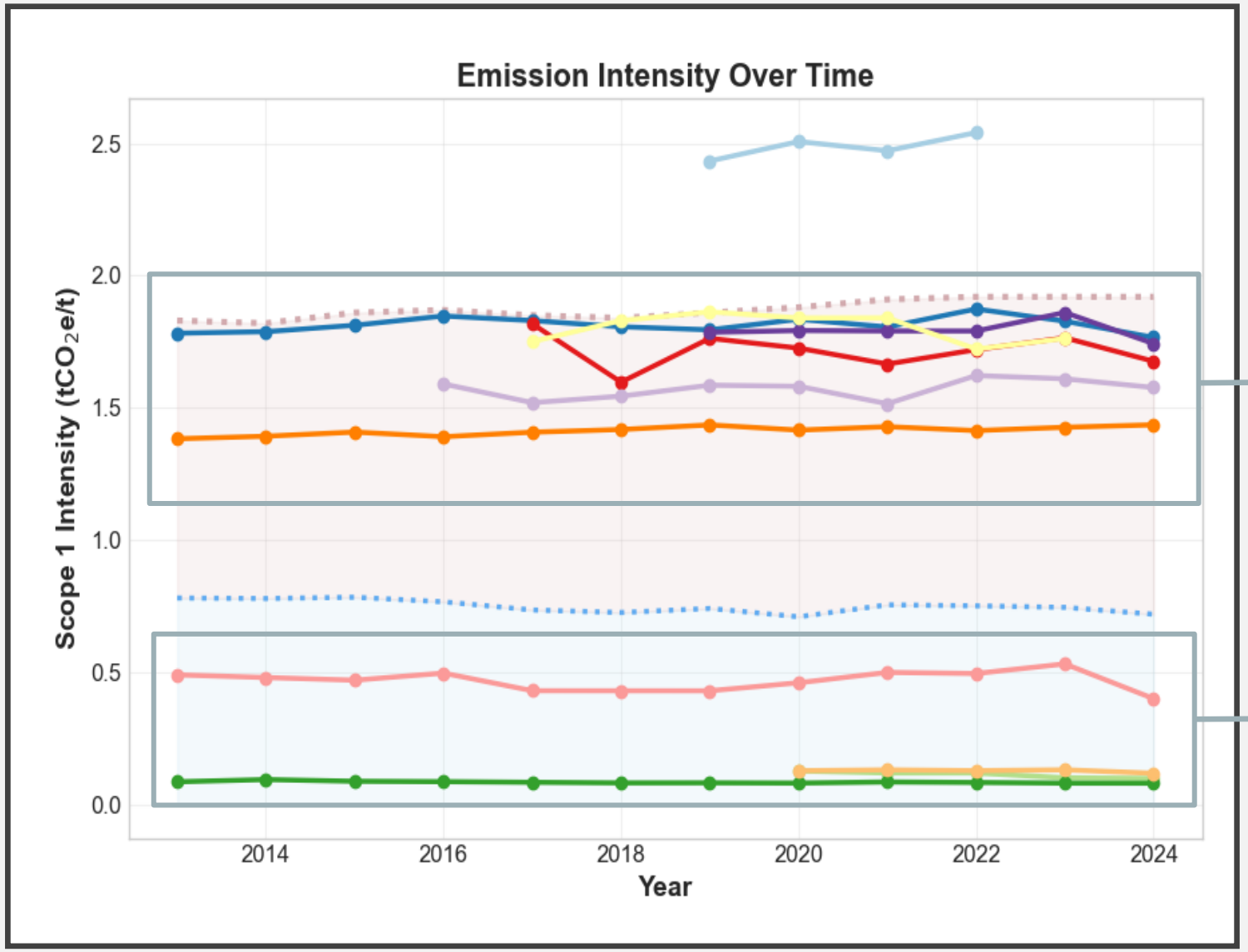
Regulates Green House Gas Emissions

Free Allocation

Technology

Panel OLS

Policies	Pricing	ETS
no significant influence	no significant influence	not meaningful



BF-BOF

Blast Furnace

Basic Oxygen Steelmaking

Coal & Iron Ore

1.8 tCo_2e/t

Replace coal - H_2

EAF

Electric Arc Furnace

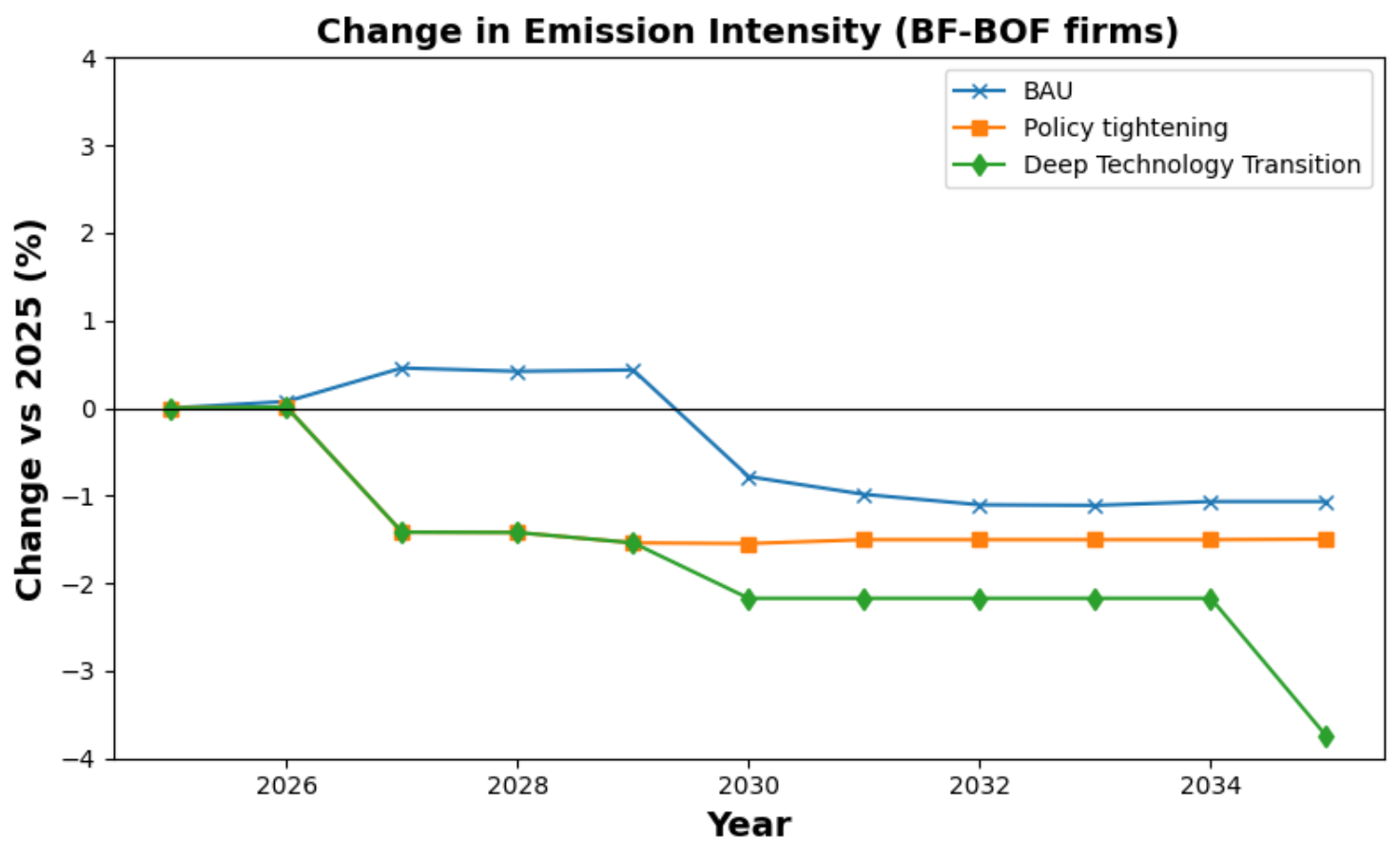
Electricity & Scrap Steel

0.3 tCo_2e/t

FUTURE ACTION (?)

Prediction through External Drivers

- **Business as usual (BAU):** current policy trends continue
- **Policy tightening:** fewer free allowances, gradual efficiency gains
- **Deep transition (hydrogen-compatible):** strict carbon regulation & structural shift in steel production



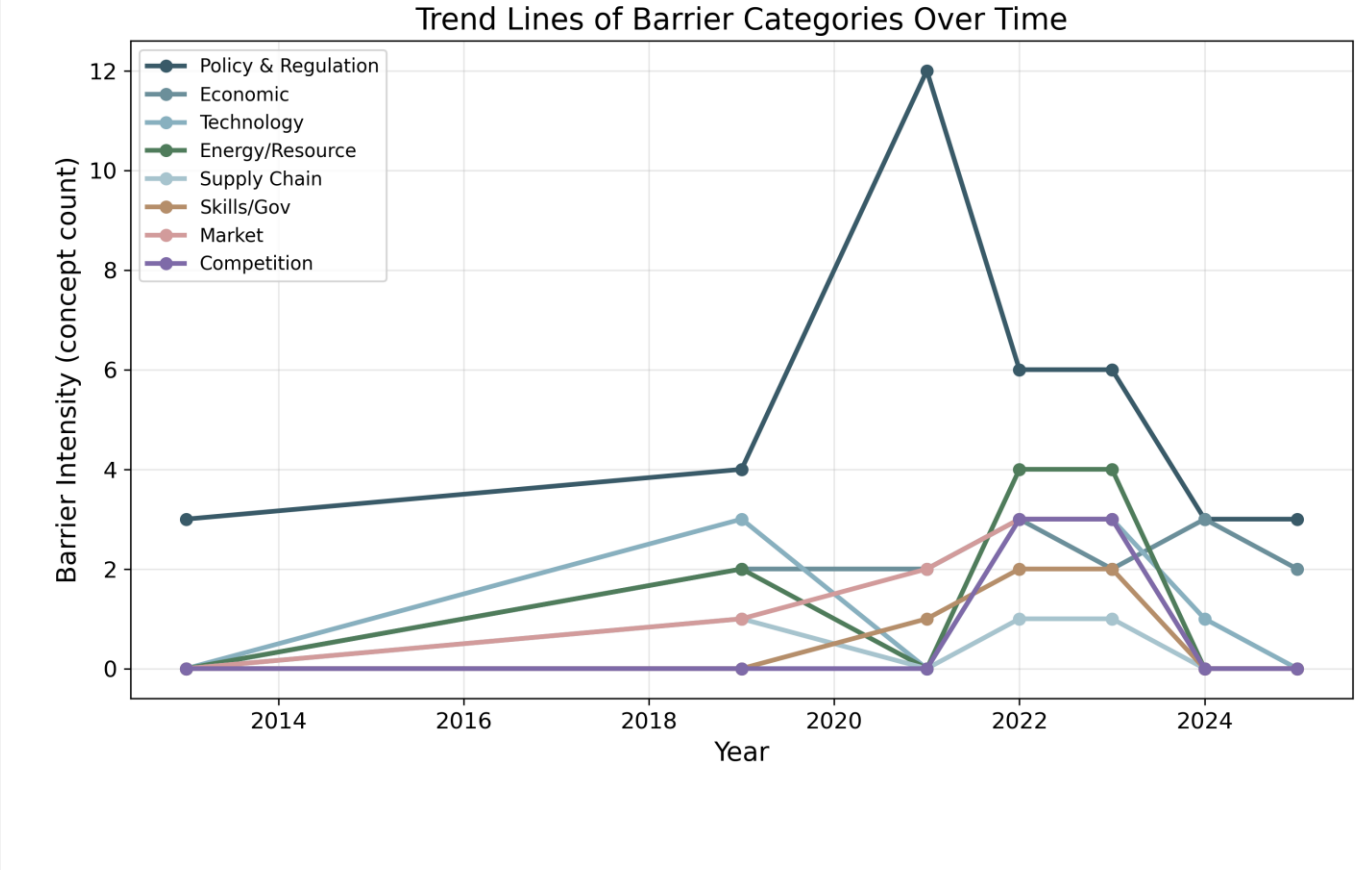
- **Panel OLS:** identifies key mission drivers & policy sensitivity trends
- **Random forest:** captures firm-level emission behavior
- **Scenarios:** altered policy and technology pathways

TALK 1

BARRIERS TO DECARBONISATION NARRATED IN REPORTS

RAG (Retrieval-Augmented Generation)

- Policy & regulation spikes in 2021
- Increasing energy/resource and technology barriers grow in 2022-23
- Emerging competition & market acceptance issues in later years

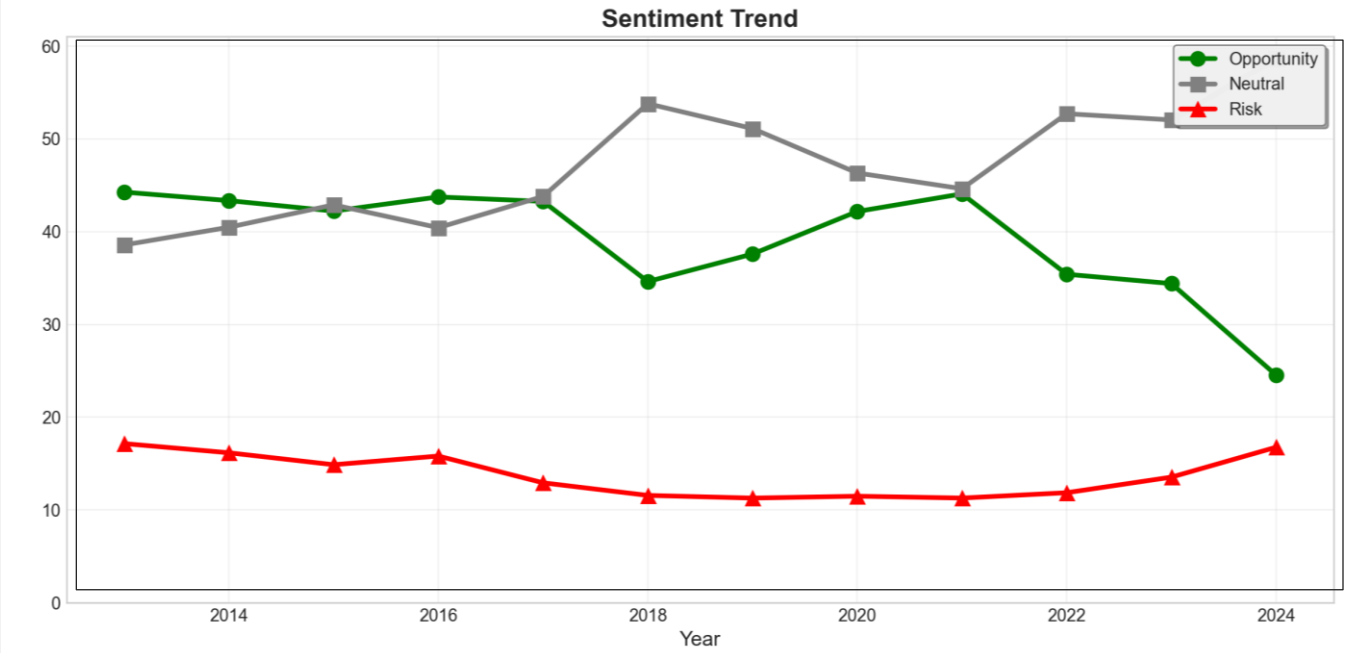


TALK II

TRENDS IN CLIMATE-RELATED SENTIMENT

ClimateBERT

- Up to 2017: Climate as business opportunity (~44%)
- 2024: More neutral/factual reporting, risk awareness returning
- Suggests: Maturing discourse, or growing concern about risks



TRANSITION TRACKER

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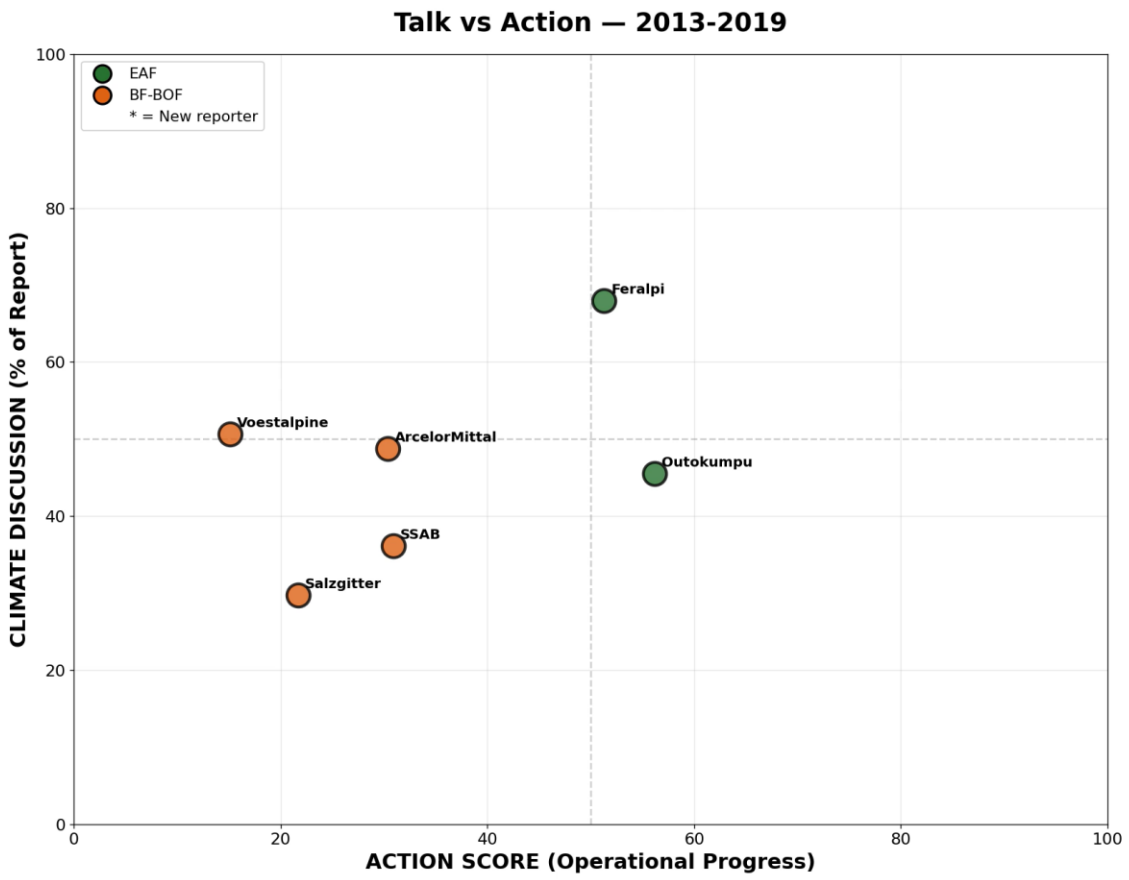
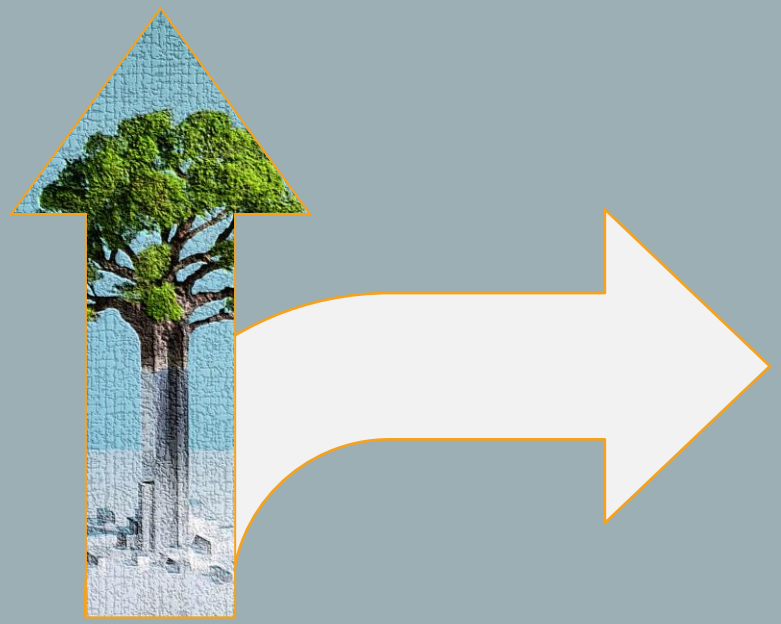
ACTION score =
Current Intensity
+ Trend
+ Data Quality
+ Future Technology
+ Energy Source

TALK score =
% text related to climate



➢ Diverse trends, changes in both directions

INSIGHTS & FUTURE WORK



Collaboration of companies and data providers needed

Public availability of data should be improved

No critical thinking without domain knowledge

Progress takes time

Streamlit

Interactive decarbonisation Transition Tracker

Stakeholders

- Steel industry,
- Policymakers,
- NGOs

Scope

Expand dataset to other regions

THANK YOU FOR YOUR ATTENTION

SuSteelAible

Aps: “I'm sus that you're steel thinking of ai rabbit holes.”

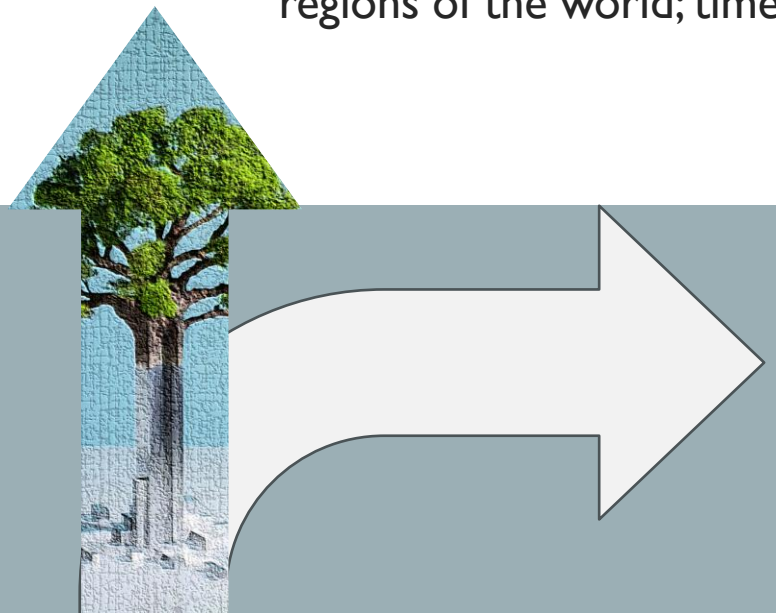
Dizzy: “I started watching math for fun…”

Karl: “A deed is worth more than a 1000 words!”

Irene: “Steel Meets Steel 🎸”

Maryam: “We overfitted the slides and underfitted the emissions.”

INSIGHTS & FUTURE WORK

- Personal insight (Reni): It's shockingly easy to lie with data
- Critical thinking is indispensable and must be paired with domain knowledge
- Despite targeted requests, neither major data-holding organisations, nor most steel producers provided CO emissions data
- Sparse, inconsistent or confidential company/plant-level data: limiting meaningful comparisons & analysis

- Streamlit page: interactive decarbonisation Transition Tracker
- Stakeholders: steel industry, policymakers, investors, NGOs
- Expanding dataset to include other regions of the world; time series analysis